RAG Basics: A Short Manual

What retrieval augmented generation is

Retrieval augmented generation (RAG) combines a search step over a trusted document collection with a large language model. Instead of relying on what the model memorised during training, the application first retrieves the passages most relevant to the user's question and then asks the model to answer using only those passages. This keeps answers grounded in documents that the operator controls and can update without retraining the model. A RAG system has two phases. The offline phase, called ingestion, reads the source files, splits them into chunks, converts each chunk into an embedding vector and stores the vectors in a vector database. The online phase, called querying, embeds the question, finds the nearest chunks and builds a prompt that contains the question together with the retrieved evidence.

Chunking and overlap

Documents are split into chunks because embedding models have a limited input length and because smaller passages produce more precise matches. A common starting point is a chunk of roughly 500 to 1000 characters. Consecutive chunks should share an overlap of about 10 to 15 percent of the chunk size, for example 90 characters for a 600 character chunk. The overlap prevents a sentence that falls on a boundary from being cut in half and lost to retrieval. Chunking should happen inside a single page when exact page citations are required. If a chunk crossed a page break, the system could not say on which page the supporting sentence appears. Very short pages produce a single chunk, while long pages produce several overlapping chunks that all keep the same page number in their metadata.

Refusal policy

A grounded assistant must be allowed to refuse. When none of the retrieved passages is close enough to the question, or when the passages do not contain the requested fact, the assistant should reply with a fixed refusal sentence rather than guessing. The refusal sentence used by this project is: I don't know based on the provided documents. Two layers implement the policy. First, a distance threshold discards weak matches before the language model is called, which also saves cost. Second, the system prompt instructs the model to use the refusal sentence whenever the remaining evidence is insufficient.